

Benzoutine Zakariae

Data scientist



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PROFILE

Data scientist with a Master's degree in Computer Science. Skilled in data analysis, machine learning, and deep learning, with hands-on experience using PyTorch and Scikit-learn.

SKILLS

Programming

Python, SQL, PyTorch, TensorFlow

Deep Learning

Model fine-tuning, CNNs, Transformers, Generative Models

Data Analysis & Visualization

Pandas, NumPy, Matplotlib, Seaborn

Computer Vision

OpenCV, Vision-Language models, Grad-CAM / attention

Deployment & Tools

Streamlit, Flask, AWS, GCP, Git, Docker

Core Competencies

Data preprocessing, Data modeling & evaluation, Neural networks, Hypothesis testing, Problem solving, Critical thinking, Domain knowledge

CERTIFICATES

Supervised Machine Learning

DeepLearning.AI

Linear Algebra for Machine Learning and Data Science

DeepLearning.AI

Data Analysis with Python

IBM

LANGUAGES

English ● ● ● ● ●
French ● ● ● ● ●
Arabic ● ● ● ● ●

PROFESSIONAL EXPERIENCES

AI Researcher - Internship

College of computing - UM6P

04/2025 – 09/2025 | Ben Guerir, Morocco

- Authored a survey paper analyzing AI applications for climate change resilience in Morocco, identifying key opportunities, challenges, and data-driven insights for future research.
- Applied hierarchical clustering on extreme precipitation events using the Toubkal supercomputer to uncover patterns in climate data.

Data scientist - Internship

Intelcap Sarl

08/2024 – 11/2024 | Rabat, Morocco

- Built a CNN model from scratch for facial emotion recognition with over 85% accuracy and applied transfer learning with ResNet50.
- Developed a sign language recognition system using OpenCV.

Machine learning | Deep learning engineer - Internship

LIMIARF - Research Laboratory

05/2024 – 09/2024 | Rabat, Morocco

- Developed Qwaze, an AI-powered platform to help teachers generate quizzes using a LLAMA 2 LLM on Amazon SageMaker.
- Deployed the model on AWS, enabling scalable and accessible quiz generation for educators.

AI developer - Internship

Province Sidi kacem

04/2023 – 06/2023 | Sidi kacem, Morocco

- Developed a document classification model with Scikit-learn and PyPDF2 for automated file categorization.
- Deployed the model via a web interface for real-time use.

PROJECTS

AI for Climate Change Resilience in Morocco

- Conducted a survey of AI research papers related to climate change resilience in Morocco.
- Used hierarchical clustering and atmospheric analysis to identify regional drivers of extreme precipitation in Morocco.

XPLAIN: A Chest X-ray Report Generator

- Built a model that generates concise radiology-style reports directly from chest X-rays.
- Achieved strong text image alignment with competitive BLEU/ROUGE scores on the OpenI dataset.

Deep Learning for Automated Medical Image Analysis: Segmentation, Tumor Detection, and Disease Classification Using the NIH Chest X-ray Dataset

- Implemented U-Net CNN for organ segmentation, automating anatomical delineation in chest X-rays.
- Built ResNet50-based model for tumor detection and disease classification, enhancing diagnostic accuracy on the NIH Chest X-ray dataset

EDUCATION

Le Wagon Data science Bootcamp

08/2025 – Present | Casablanca, Morocco

Intensive 9-week program covering data analysis, machine learning, deep learning, and deployment

Master's Degree in Computer Science and Telecommunications

Faculty of Science UM5

2023 – 2025 | Rabat, Morocco

Bachelor's degree in Networking and Telecommunications

Faculty of Science UM5

2022 – 2023 | Rabat, Morocco

Diploma BTS in Computer Systems and Networks

AlFarabi Technical High School

2020 – 2022 | Salé, Morocco